

Finite/Fixed-Time Pinning Synchronization of Complex Networks With Stochastic Disturbances

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Abstract—This brief proposes a unified theoretical framework to investigate the finite/fixed-time synchronization of complex networks with stochastic disturbances. By designing a common pinning controller with different ranges of power parameters, both the goals of finite-time and fixed-time synchronization in probability for the network topology containing spanning trees can be achieved. Moreover, with the help of finite-time stochastic stability theory, two types of explicit expressions of finite/fixed (dependent/independent on the initial values) settling times are calculated as well. One numerical example is finally presented to demonstrate the effectiveness of the theoretical analysis.

Index Terms—Complex networks, finite-time synchronization, fixed-time synchronization, pinning controllers.

I. INTRODUCTION

Recently, finite-time synchronization has received increasing attention due to its fast convergence rate and disturbance rejection ability [1]–[5]. It is known that rapid convergence can exactly disclose how long it will take to achieve the synchronization [6]–[8]. However, one critical problem which should be noticed is that the settling time of finite-time control is heavily dependent on the initial conditions, and different initial conditions may lead to different convergence time [9]. Especially for a large system in practical applications, it is hard to adjust or estimate the initial conditions, which will directly result in the inaccessibility of the control time and then declining of the systems performance [10].

To amend the above-mentioned drawbacks of finite-time control, a new concept called fixed-time convergence is proposed by Polyakov [10]. Thus, it ensures the finite-time convergence, and guarantees that the setting-time is uniformly bounded for any initial value. Following this idea, further investigations of fixed-time control problems have been presented in [11]–[14], where the relationship between the finite-time and fixed-time convergence was revealed from different aspects. However, such results were obtained based on some

complicated calculation formula and restrictive conditions. It is significant and challenging to recover the essence of finite-time and fixed-time convergence by a simple and unified approach.

In addition to the finite settling time, the disturbance rejection is another potential ability of finite-time synchronization. For example, Shen *et al.* [15] considered the issues of finite-time synchronization for Markov jump neural networks by a parameter-dependent Lyapunov functional. Liu *et al.* [16] investigated the finite-time consensus problems of second-order disturbed multiagent systems by proposing a finite-time convergent discontinuous observer. However, only deterministic disturbances were considered in the above references.

Actually, the evolution of many practical systems is always affected by various stochastic disturbances and uncertainties from the unpredictable environmental conditions. To this point, Yin *et al.* [17], [18] studied the issue of almost surely finite-time stabilization for a class of stochastic nonlinear systems by proposing a new finite-time stability theorem. Zhang *et al.* [19] further solved the problem of fixed-time synchronization for complex networks under stochastic perturbations. However, the finite-time synchronization or the fixed-time one for complex networks with stochastic disturbances are always discussed separately. The question then arises, does there exist a unified framework to study both of them by a common control protocol?

Motivated by the above concerns, this brief further investigates the finite/fixed-time synchronization (FFTS) problems of complex networks with stochastic disturbances. The main results of this brief extend the existing works in the following aspects.

- 1) Compared with the finite-time and fixed-time control studied separately, the goal of FFTS can be achieved under a common controller, and the settling time is either finite or fixed can be decided by regulating power parameters.
- 2) The new designed fixed-time synchronization controller can effectively eliminate the dependence on the initial values of finite-time synchronization. Also, it can successfully reject the stochastic disturbances of complex networks.
- 3) In order to reduce the high control cost in a large-scale network, the FFTS issue is studied by a pinning control strategy. Thus, it can efficiently drive the network to a desired synchronous state by applying local control actions to a small fraction of network nodes.
- 4) With the help of the finite-time stability in probability theory, the explicit expressions of finite/fixed (dependent/independent on the initial values) settling times are estimated, respectively.

The notations in this brief are standard. $\text{diag}(a_i)$ ($i = 1, \dots, n$) denotes a block-diagonal matrix with the main diagonal entries $a_1, \dots, a_n$; $\lambda_1(\cdot)$, $\lambda_2(\cdot)$, and $\lambda_{\max}(\cdot)$ denote the minimum, second minimum, and maximum eigenvalues of a real symmetric matrix, respectively. $\text{sign}(\cdot)$ is the sign function with $\text{sign}(0) = 0$ and $\text{sgn}^k(\cdot) = \text{sign}(\cdot) \cdot |\cdot|^k$. A function $V: \mathbb{R}^n \rightarrow \mathbb{R}$ is said to be C_k^0 if it is k -times continuously differentiable. A function $\mu: \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is said to be a class K function if it is continuous, strictly increasing and $\mu(0) = 0$. A class K function μ is said to belong to class K_∞ if $\mu(r) \rightarrow \infty$ as $r \rightarrow \infty$.

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II. PRELIMINARIES AND MODEL FORMULATION

Consider a complex network composed of N nodes as the following differential equations:

$$dx_i(t) = (f(x_i(t)) + u_i(t))dt + p_i(t, x_i(t))d\omega_i(t) \quad (1)$$

where $x_i(t) = (x_{i1}(t), x_{i2}(t), \dots, x_{in}(t))^T \in \mathbb{R}^n$ ($i \in \{1, 2, \dots, N\}$) represents the state variables of i th node at time t . Denote $f(\cdot)$ to be the intrinsic dynamics of the node. Let u_i represent the distributed coupling controller that to be designed. $\omega_i(t) = (\omega_{i1}(t), \omega_{i2}(t), \dots, \omega_{in}(t))^T \in \mathbb{R}^n$ is a bounded vector-form Wiener process (Brownian motion) on the probability space $(\Omega, \mathcal{F}, \mathcal{P})$ satisfying $\mathbb{E}\{d\omega_{ij}(t)\} = 0$ and $\mathbb{E}\{d\omega_{ij}^2(t)\} = dt$. $p_i(\cdot, \cdot) : \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^{n \times n}$ is the diffusive coupling matrix representing noisy intensity, which satisfies the following assumption.

Assumption 1: For $\forall x, y \in \mathbb{R}^n$, there exist positive constants ψ_i such that

$$\begin{aligned} & \text{trace}[(p_i(t, x) - p_i(t, y))^T (p_i(t, x) - p_i(t, y))] \\ & \leq \psi_i(x - y)^T (x - y), i \in \mathcal{N}. \end{aligned} \quad (2)$$

Our control objective is to finite/fixed-timely synchronize complex network (1) to a desired reference state $x_0(t) \in \mathbb{R}^n$, which is a solution of an isolated node and satisfies

$$dx_0(t) = f(x_0(t))dt. \quad (3)$$

Generally, $x_0(t)$ can be an equilibrium point or a periodic/aperiodic/chaotic orbit in the phase space and is noise-free with $p_i(t, x_0(t)) = 0_{n \times n}$ [19]–[21].

The nodes dynamics $f(\cdot)$ satisfies the usual Lipschitz condition.

Assumption 2: For $\forall x, y \in \mathbb{R}^n$, there exists a positive constant ρ such that

$$[x - y]^T [f(x) - f(y)] \leq \rho [x - y]^T [x - y]. \quad (4)$$

It is known that the existence of solutions for the complex network (1) and (3) can be guaranteed under the above two assumptions [22], [23].

The communication topology $\mathcal{G}$ considered in this brief contains spanning trees and each strongly connected component of the topology is detail-balanced (that is, there exist some positive scalars q_i , such that $q_i a_{ij} = q_j a_{ji}$ for all $i, j \in \mathcal{N}$, where $A = [a_{ij}]_{\mathcal{N} \times \mathcal{N}}$ represents the weighted adjacency matrix). The Laplacian matrix with respect to graph $\mathcal{G}$ is denoted as L_A . With the help of [17, Definition 3.1], our goal is to design the control protocol u_i such that all the nodes x_i can synchronize to the desired state x_0 in probability within a finite/fixed time, just as claimed in the following two definitions.

Definition 1: The complex network with dynamics (1) and (3) is said to achieve finite-time synchronization in probability if there exists a time $T_{x(0)}$, such that $P\{\lim_{t \rightarrow T_{x(0)}} x_i(t) = x_0(t)\} = 1, \forall i \in \mathcal{N}$.

Definition 2: The complex network with dynamics (1) and (3) is said to achieve fixed-time synchronization in probability if the finite-time synchronization is solved and the stochastic settling-time function $T_{x(0)}$ is further uniformly bounded by a fixed positive constant.

Lemma 1 [24]: Let $a_1, a_2, \dots, a_n$ be positive numbers. If $0 < k < l$, then

$$\left(\sum_{i=1}^n a_i^l\right)^{1/l} \leq \left(\sum_{i=1}^n a_i^k\right)^{1/k} \leq n^{\frac{1}{k}-\frac{1}{l}} \left(\sum_{i=1}^n a_i^l\right)^{1/l}.$$

III. MAIN RESULTS

This section formulates some basic results on the issue of FFTS for the disturbed complex networks under a unified pinning protocol.

Lemma 2 [18]: For the continuous stochastic nonlinear system

$$dx(t) = g(x(t))dt + h(x(t))d\omega(t) \quad (5)$$

with $g(0) = h(0) = 0$, if there exists a $C_2^0(\mathbb{R}^n)$ function $V : \mathbb{R}^n \rightarrow \mathbb{R}^+$, two K_∞ class functions κ_1, κ_2 , and two positive constants γ, α with $0 < \alpha < 1$, such that for all $t \geq 0$

$$\kappa_1(x) \leq V(x) \leq \kappa_2(x), x \in \mathbb{R}^n \quad (6)$$

$$\mathcal{L}V(x) \leq -\gamma V^\alpha(x), x \in \mathbb{R}^n \setminus \{0\} \quad (7)$$

then the trivial zero solution of (5) is finite-time stable in probability and the stochastic settling time satisfies $\mathbb{E}\{T_{x(0)}\} \leq ([V(x(0))]^{1-\alpha})/[\gamma(1-\alpha)]$, where $\mathcal{L}V = (\partial V(x)/\partial x)g(x) + (1/2)\text{trace}\{h^T(x)(\partial^2 V(x)/\partial x^2)h(x)\}$.

Remark 1: If the above inequality (7) changes to

$$\mathcal{L}V(x) \leq -\gamma_1 V^\alpha(x) - \gamma_2 V^\beta(x), x \in \mathbb{R}^n \setminus \{0\} \quad (8)$$

and other conditions in Lemma 2 remain unchanged, then based on the fixed-time stability results in [10], [11], [13], and [14], the origin of (5) is fixed-time stable in probability, where $\gamma_1, \gamma_2, \alpha$, and β are positive constants with $0 < \alpha < 1$ and $\beta > 1$.

First, we assume that the communication topology $\mathcal{G}$ is strongly connected and detail-balanced. With the help of pinning control method [25], the controller $u_i(t)$ is designed as

$$\begin{aligned} u_i(t) = & \sum_{j=1}^N q_i a_{ij} [\zeta \text{sgn}^\alpha(x_j - x_i) + \eta \text{sgn}^\beta(x_j - x_i)] \\ & - b_i \text{sgn}^\gamma(x_i - x_0) \end{aligned} \quad (9)$$

where control gains $\zeta, \eta > 0$; the pinning gains $b_i \geq 0$ ($b_i > 0$ for the pinned nodes); $q_i > 0$ satisfies $q_i a_{ij} = q_j a_{ji}$. α, β , and γ are three positive power parameters that can decide the setting time to be finite or fixed according to the following theorem.

Theorem 1: For the strongly connected and detail-balanced network topology with Assumptions 1, 2, and the pinning controller (9) with $0 < \gamma = \alpha < 1$.

- 1) When $\beta < 1$, finite-time synchronization in probability can be achieved if $\lambda_3 < \min\{\zeta[\lambda_1(\mathbf{A})]^{(\alpha+1)/2}/\|e(0)\|^{1-\alpha}, \eta[2\lambda_2(\bar{L}_{\bar{A}})]^{(\beta+1)/2}/\|e(0)\|^{1-\beta}\}$.
- 2) When $\beta > 1$, fixed-time synchronization in probability can be achieved if $\lambda_3 < \min\{\zeta[\lambda_1(\mathbf{A})]^{(\alpha+1)/2}, \eta N_\beta[2\lambda_2(\bar{L}_{\bar{A}})]^{(\beta+1)/2}\}$, where $\mathbf{A} = 2L_{\bar{A}} + \tilde{\mathbf{b}}$ and $\bar{A} = [q_i a_{ij}]_{\mathcal{N} \times \mathcal{N}}^{[2/(\alpha+1)]}$, $\tilde{\mathbf{b}} = \text{diag}[(2b_i/\zeta)^{2/(\alpha+1)}]$; $\bar{A} = [q_i a_{ij}]_{\mathcal{N} \times \mathcal{N}}^{[2/(\beta+1)]}$; $N_\beta = n^{[(1-\beta)/2]}N^{1-\beta}$; $\lambda_3 = 2\rho + \max_{1 \leq i \leq N}\{\psi_i\}$; $e(0) = (e_1^T(0), \dots, e_N^T(0))^T$, and $e_i(0) = x_i(0) - x_0(0)$, $i \in \mathcal{N}$.

Proof: Subtracting (3) from (1) with (9), we obtain the error dynamical system

$$\begin{aligned} de_i(t) = & \left\{ \tilde{f}(e_i) + \zeta \sum_{j=1}^N q_i a_{ij} \text{sgn}^\alpha(e_j - e_i) \right. \\ & + \eta \sum_{j=1}^N q_i a_{ij} \text{sgn}^\beta(e_j - e_i) - b_i \text{sgn}^\gamma(e_i) \left. \right\} dt \\ & + \tilde{p}_i(t, e_i) d\omega_i(t) \end{aligned} \quad (10)$$

where $e_i(t) = x_i(t) - x_0(t)$, $\tilde{f}(e_i) = f(x_i) - f(x_0)$, and $\tilde{p}_i(t, e_i) = p_i(t, x_i) - p_i(t, x_0)$.

Now, the issue of FFTS for complex network (1) is transformed into the problem of finite/fixed-time stability of the error system (10). To this end, we choose the Lyapunov function as $V(t) = \sum_{i=1}^N e_i^T(t) e_i(t)$. Based on the Itô's formula, the time derivative of $V(t)$ along the trajectories of system (10) is calculated as the stochastic differential

$$dV(t) = \mathcal{L}V(t)dt + 2 \sum_{i=1}^N e_i^T(t) \tilde{p}_i(t, e_i(t)) d\omega_i(t) \quad (11)$$

where the differential operator

$$\begin{aligned}\mathcal{L}V(t) &= \frac{\partial V}{\partial e}\delta(t) + \frac{1}{2}\text{trace}\left\{\tilde{p}^T(t)\frac{\partial^2 V}{\partial e^2}\tilde{p}(t)\right\} \\ &= 2\sum_{i=1}^N e_i^T(t)\left\{\tilde{f}(e_i(t)) + \sum_{j=1}^N q_{iaij}[\zeta\text{sgn}^\alpha(e_j(t) - e_i(t)) + \eta\text{sgn}^\beta(e_j(t) - e_i(t)) - b_i\text{sgn}^\gamma(e_i(t))]\right\} \\ &\quad + \text{trace}\left[\tilde{p}_i^T(t, e_i(t))\tilde{p}_i(t, e_i(t))\right]\end{aligned}\quad (12)$$

and $\delta(t) = (\delta_1^T(t), \delta_2^T(t), \dots, \delta_N^T(t))^T$, $\delta_i(t) = \tilde{f}(e_i) + \zeta\sum_{j=1}^N q_{iaij}\text{sgn}^\alpha(e_j - e_i) + \eta\sum_{j=1}^N q_{iaij}\text{sgn}^\beta(e_j - e_i) - b_i\text{sgn}^\gamma(e_i)$, $e(t) = (e_1^T(t), e_2^T(t), \dots, e_N^T(t))^T$, $\tilde{p}(t) = (\tilde{p}_1(t, e_1), \tilde{p}_2(t, e_2), \dots, \tilde{p}_N(t, e_N))^T$.

Noticing the fact that $q_{iaij} = q_{ajji}$, $\forall i, j \in \mathcal{N}$, we can obtain

$$\begin{aligned}\sum_{i,j=1}^N q_{iaij}e_i^T &\left[\zeta\text{sgn}^\alpha(e_j - e_i) + \eta\text{sgn}^\beta(e_j - e_i)\right] \\ &= \sum_{i,j=1}^N q_{ajji}e_j^T \left[\zeta\text{sgn}^\alpha(e_i - e_j) + \eta\text{sgn}^\beta(e_i - e_j)\right] \\ &= -\frac{1}{2}\sum_{i,j=1}^N q_{iaij}(e_j - e_i)^T \left[\zeta\text{sgn}^\alpha(e_j - e_i) + \eta\text{sgn}^\beta(e_j - e_i)\right] \\ &= -\frac{\zeta}{2}\sum_{i,j=1}^N \sum_{k=1}^n q_{iaij}|e_{jk} - e_{ik}|^{\alpha+1} \\ &\quad - \frac{\eta}{2}\sum_{i,j=1}^N \sum_{k=1}^n q_{iaij}|e_{jk} - e_{ik}|^{\beta+1}\end{aligned}\quad (13)$$

where the last equality is obtained due to the results in [26] and [27].

Combing the Assumptions 1 and 2 with the above formula (13), we can yield

$$\begin{aligned}\mathcal{L}V(t) &\leq 2\rho\sum_{i=1}^N e_i^T(t)e_i(t) \\ &\quad - \zeta\sum_{i,j=1}^N \sum_{k=1}^n q_{iaij}|e_{jk} - e_{ik}|^{\alpha+1} \\ &\quad - \eta\sum_{i,j=1}^N \sum_{k=1}^n q_{iaij}|e_{jk} - e_{ik}|^{\beta+1} \\ &\quad - 2\sum_{i,j=1}^N b_i\sum_{k=1}^n |e_{ik}|^{\gamma+1} + \sum_{i=1}^N \psi_i e_i^T(t)e_i(t).\end{aligned}\quad (14)$$

In the following, three terms in (14) with different power parameters α , β , and γ will be discussed, respectively. They are the key parameters to achieve the finite/fixed-time stability of error system (10).

First, for the parameters α and γ , due to $0 < \gamma = \alpha < 1$, Lemma 1 implies that

$$\begin{aligned}&-\zeta\sum_{i,j=1}^N \sum_{k=1}^n q_{iaij}|e_{jk} - e_{ik}|^{\alpha+1} \\ &\quad - 2\sum_{i=1}^N b_i\sum_{k=1}^n |e_{ik}|^{\gamma+1} \\ &= -\zeta\sum_{k=1}^n \left[\sum_{i,j=1}^N q_{iaij}|e_{jk} - e_{ik}|^{\alpha+1} + \sum_{i=1}^N \frac{2b_i}{\zeta}|e_{ik}(t)|^{\alpha+1}\right] \\ &\leq -\zeta\left[\sum_{k=1}^n \left(\sum_{i,j=1}^N (q_{iaij})^{\frac{2}{\alpha+1}}|e_{jk} - e_{ik}|^2\right)^{\frac{\alpha+1}{2}} + \sum_{i=1}^N \left(\frac{2b_i}{\zeta}\right)^{\frac{2}{\alpha+1}}|e_{ik}|^2\right]^{\frac{\alpha+1}{2}}\end{aligned}$$

$$\begin{aligned}&= -\zeta\left[\sum_{i=1}^N e_i^T(t)\left[2L_{\tilde{A}} + \tilde{\mathbf{b}}\right]e_i(t)\right]^{\frac{\alpha+1}{2}} \\ &= -\zeta\left[e^T(t)(\mathbf{A} \otimes I_n)e(t)\right]^{\frac{\alpha+1}{2}} \\ &\leq -\zeta[\lambda_1(\mathbf{A})]^{\frac{\alpha+1}{2}}\left[e^T(t)e(t)\right]^{\frac{\alpha+1}{2}} \\ &= -\zeta[\lambda_1(\mathbf{A})]^{\frac{\alpha+1}{2}}[V(t)]^{\frac{\alpha+1}{2}}\end{aligned}\quad (15)$$

where $\lambda_1(\mathbf{A}) > 0$ based on the results in [26] and [27].

Next, for the parameter β , it is claimed that the synchronization time being finite or fixed is up to the ranges of power parameter β , which will be explained in detail below.

1) If $\beta < 1$, it is deduced by Lemma 1 that

$$\begin{aligned}&-\eta\sum_{i,j=1}^N \sum_{k=1}^n q_{iaij}|e_{jk} - e_{ik}|^{\beta+1} \\ &\leq -\eta\left[\sum_{k=1}^n \sum_{i,j=1}^N (q_{iaij})^{\frac{2}{\beta+1}}|e_{jk} - e_{ik}|^2\right]^{\frac{\beta+1}{2}} \\ &= -\eta\left[2\sum_{i=1}^N e_i^T(t)L_{\tilde{A}}e_i(t)\right]^{\frac{\beta+1}{2}} \\ &= -\eta[2e^T(t)(L_{\tilde{A}} \otimes I_n)e(t)]^{\frac{\beta+1}{2}} \\ &\leq -\eta[2\lambda_2(L_{\tilde{A}})]^{\frac{\beta+1}{2}}[V(t)]^{\frac{\beta+1}{2}}\end{aligned}\quad (16)$$

where $\lambda_2(L_{\tilde{A}}) > 0$ [26], [27].

Combining (14) and (15) with (16), we can further get

$$\begin{aligned}\mathcal{L}V(t) &\leq -\zeta[\lambda_1(\mathbf{A})]^{\frac{\alpha+1}{2}}[V(t)]^{\frac{\alpha+1}{2}} \\ &\quad - \eta[2\lambda_2(L_{\tilde{A}})]^{\frac{\beta+1}{2}}[V(t)]^{\frac{\beta+1}{2}} \\ &\quad + 2\rho V(t) + \max_{1 \leq i \leq N} \{\psi_i\} \cdot V(t) \\ &= -\zeta[\lambda_1(\mathbf{A})]^{\frac{\alpha+1}{2}}[V(t)]^{\frac{\alpha+1}{2}} \\ &\quad - \eta[2\lambda_2(L_{\tilde{A}})]^{\frac{\beta+1}{2}}[V(t)]^{\frac{\beta+1}{2}} \\ &\quad + \lambda_3 V(t).\end{aligned}\quad (17)$$

Taking the mathematical expectation of both sides of (11) with (17), we have

$$\mathbb{E}[dV(t)] \leq \lambda_3\mathbb{E}[V] - \bar{\zeta}\mathbb{E}[V]^{\frac{\alpha+1}{2}} - \bar{\eta}\mathbb{E}[V]^{\frac{\beta+1}{2}}\quad (18)$$

where $\bar{\zeta} = \zeta[\lambda_1(\mathbf{A})]^{[(\alpha+1)/2]}$, $\bar{\eta} = \eta[2\lambda_2(L_{\tilde{A}})]^{[(\beta+1)/2]}$.

Then, owing to $[(\alpha+1)/2], [(\beta+1)/2] < 1$, Lemma 2 can ensure the error system (10) is finite-time stable in probability. That is, there exists a stochastic time T_1 , such that $\mathbb{E}\{V(t)\} = 0$ for all $t \geq T_1$, which also implies that $P\{x_i = x_0\} = 1$ ($\forall i \in \mathcal{N}$) in finite time. Actually, with the help of [28, Lemma 1], the settling time T_1 can be estimated as

$$\begin{aligned}\mathbb{E}\{T_1\} &\leq \int_0^{V(0)} \frac{dV}{-\lambda_3 V + \bar{\zeta} V^{\frac{\alpha+1}{2}} + \bar{\eta} V^{\frac{\beta+1}{2}}} \\ &\leq \min \left\{ \int_0^{V(0)} \frac{dV}{-\lambda_3 V + \bar{\zeta} V^{\frac{\alpha+1}{2}}}, \int_0^{V(0)} \frac{dV}{-\lambda_3 V + \bar{\eta} V^{\frac{\beta+1}{2}}} \right\} \\ &= \min \left\{ \frac{2}{\lambda_3(1-\alpha)} \ln \frac{\bar{\zeta}}{\bar{\zeta} - \lambda_3 e_\alpha(0)}, \frac{2}{\lambda_3(1-\beta)} \ln \frac{\bar{\eta}}{\bar{\eta} - \lambda_3 e_\beta(0)} \right\}\end{aligned}\quad (19)$$

which is heavily dependent on the initial states.

2) If $\beta > 1$, then Lemma 1 leads to

$$\begin{aligned}
& -\eta \sum_{i,j=1}^N \sum_{k=1}^n q_{ij} a_{ij} |e_{jk} - e_{ik}|^{\beta+1} \\
& \leq -\eta n^{\frac{1-\beta}{2}} N^{1-\beta} \left[\sum_{k=1}^n \sum_{i,j=1}^N (q_{ij} a_{ij})^{\frac{2}{\beta+1}} |e_{jk} - e_{ik}|^2 \right]^{\frac{\beta+1}{2}} \\
& = -\eta N_\beta \left[2 \sum_{i=1}^N e_i^T(t) L_{\bar{A}} e_i(t) \right]^{\frac{\beta+1}{2}} \\
& = -\eta N_\beta \left[2e^T(t) (L_{\bar{A}} \otimes I_n) e(t) \right]^{\frac{\beta+1}{2}} \\
& \leq -\eta N_\beta [2\lambda_2(L_{\bar{A}})]^{\frac{\beta+1}{2}} [V(t)]^{\frac{\beta+1}{2}}. \tag{20}
\end{aligned}$$

Combining (14) and (15) with (20), we get

$$\mathcal{L}V(t) \leq -\bar{\zeta}[V(t)]^{\frac{\alpha+1}{2}} - \eta N_\beta [V(t)]^{\frac{\beta+1}{2}} + \lambda_3 V(t). \tag{21}$$

Now, following the similar line of case 1 and taking the mathematical expectation of both sides of (11) with (21), we can obtain

$$\mathbb{E}[dV(t)] \leq \lambda_3 \mathbb{E}[V] - \bar{\zeta} \mathbb{E}[V]^{\frac{\alpha+1}{2}} - \eta N_\beta \mathbb{E}[V]^{\frac{\beta+1}{2}}. \tag{22}$$

Then, due to $0 < [(\alpha+1)/2] < 1 < [(\beta+1)/2]$, it is claimed from Lemma 2 and Remark 1 that the error system (10) is fixed-time stable in probability. That is, there exists a finite value T_2 , such that $P\{\lim_{t \rightarrow T_2} x_i(t) = x_0(t)\} = 1, \forall i \in \mathcal{N}$, and T_2 is uniformly bounded. In fact, the settling time T_2 can be estimated as

$$\begin{aligned}
\mathbb{E}\{T_2\} & \leq \int_0^{V(0)} \frac{dV}{-\lambda_3 V + \bar{\zeta} V^{\frac{\alpha+1}{2}} + \eta N_\beta V^{\frac{\beta+1}{2}}} \\
& = \int_0^{V(0)} \frac{dV}{V^{\frac{\alpha+1}{2}} \left(\eta N_\beta V^{\frac{\beta-\alpha}{2}} + \bar{\zeta} - \lambda_3 V^{\frac{1-\alpha}{2}} \right)} \\
& = \frac{2}{1-\alpha} \int_0^{V(0)} \frac{dV^{\frac{1-\alpha}{2}}}{\eta N_\beta V^{\frac{\beta-\alpha}{2}} + \bar{\zeta} - \lambda_3 V^{\frac{1-\alpha}{2}}}. \tag{23}
\end{aligned}$$

Denote $z = V^{(1-\alpha)/2}$, then

$$\mathbb{E}\{T_2\} \leq \frac{2}{1-\alpha} \int_1^{V(0)^{\frac{1-\alpha}{2}}} \frac{dz}{\eta N_\beta z^{1+\frac{\beta-1}{1-\alpha}} + \bar{\zeta} - \lambda_3 z}. \tag{24}$$

If $V(0)^{(1-\alpha)/2} \leq 1$, then

$$\begin{aligned}
\mathbb{E}\{T_2\} & \leq \frac{2}{1-\alpha} \int_0^1 \frac{dz}{\eta N_\beta z^{1+\frac{\beta-1}{1-\alpha}} + \bar{\zeta} - \lambda_3 z} \\
& \leq \frac{2}{1-\alpha} \int_0^1 \frac{dz}{\bar{\zeta} - \lambda_3 z} \\
& = \frac{2}{1-\alpha} \frac{1}{\bar{\zeta} - \lambda_3}. \tag{25}
\end{aligned}$$

If $V(0)^{(1-\alpha)/2} > 1$, then

$$\begin{aligned}
\mathbb{E}\{T_2\} & \leq \frac{2}{1-\alpha} \int_0^1 \frac{dz}{\eta N_\beta z^{1+\frac{\beta-1}{1-\alpha}} + \bar{\zeta} - \lambda_3 z} \\
& + \frac{2}{1-\alpha} \int_1^{V(0)^{\frac{1-\alpha}{2}}} \frac{dz}{\eta N_\beta z^{1+\frac{\beta-1}{1-\alpha}} + \bar{\zeta} - \lambda_3 z} \\
& \leq \frac{2}{1-\alpha} \int_0^1 \frac{dz}{\bar{\zeta} - \lambda_3 z} \\
& + \frac{2}{1-\alpha} \int_1^{V(0)^{\frac{1-\alpha}{2}}} \frac{dz}{(\eta N_\beta - \lambda_3) z^{1+\frac{\beta-1}{1-\alpha}}}
\end{aligned}$$

$$\begin{aligned}
& = \frac{2}{1-\alpha} \frac{1}{\bar{\zeta} - \lambda_3} + \frac{1}{\eta N_\beta - \lambda_3} \left[\frac{2}{\beta-1} - \frac{2V(0)^{\frac{1-\beta}{2}}}{\beta-1} \right] \\
& \leq \frac{2}{1-\alpha} \frac{1}{\bar{\zeta} - \lambda_3} + \frac{2}{\beta-1} \frac{1}{\eta N_\beta - \lambda_3}. \tag{26}
\end{aligned}$$

To sum up, we have $\mathbb{E}\{T_2\} \leq [2/(1-\alpha)][1/(\bar{\zeta} - \lambda_3)] + [2/(\beta-1)][1/(\eta N_\beta - \lambda_3)]$, which is indeed a fixed value independent of any initial value. This completes the proof. ■

Remark 2: According to Theorem 1, both the finite-time and fixed-time synchronization goals are achieved by a unified pinning controller with different power parameters. Here, α and γ belong to interval $(0, 1)$ and different ranges of power β result in different synchronization time. In particular, when $\beta < 1$, the settling time T_1 in (19) is mainly dependent on the initial values, which is estimated with the help of [28, Lemma 1]. However, such dependence will vanish once $\beta > 1$, and fixed-time synchronization is then realized. The fixed settling time T_2 is calculated directly by Theorem 1, which is also consistent with the results in [12]–[14].

Remark 3: Without loss of generality, the power γ in the pinning term can be assumed to be equal to the power α in Theorem 1. In fact, γ can also be chosen as the same as β , and the pinning term $b_i \text{sgn}^\gamma(x_i - x_0)$ can be incorporated into the term of $\eta \sum_{j=1}^N q_{ij} a_{ij} \text{sgn}^\beta(x_j - x_i)$. Hence, (15), (16), and (20) should be updated accordingly. The whole proof of letting power γ be equal to β in the pinning term, is similar to that of Theorem 1 and hence is omitted here.

Remark 4: If let $b_i \equiv 0$ ($i \in \mathcal{N}$) in the synchronization controller $u_i(t)$ in (9), then the corresponding fixed-time control results can be included by Theorem 1 as a special case. Hence, Theorem 1 can be viewed as an extension of the results obtained in [29] by generalizing the control protocols and the control objectives. Moreover, Theorem 1 extends the results in [15] and [16] to discuss the case of fixed-time synchronization and analyze how the power parameters affect the convergence time and study how to handle with the stochastic external disturbances.

Now, for the general network topology containing spanning trees, we have the following main results.

Theorem 2: For the network topology $\mathcal{G}$ containing spanning trees and each strongly connected component of the topology is detail-balanced, under the assumptions of Theorem 1, the finite-time and fixed-time synchronization in probability can be achieved, respectively.

Proof: Without loss of generality, it is supposed that the network topology contains two strongly connected components (other cases still can be solved by induction), which are denoted as $Q_1 = \{1, 2, \dots, s_1\}$ for the leader node set and $Q_2 = \{s_1+1, s_1+2, \dots, N\}$ for the follower node one. Because both of them are detail-balanced, then there exist $(\hat{q}_1, \dots, \hat{q}_{s_1})^T$ and $(\hat{q}_{s_1+1}, \dots, \hat{q}_N)^T$ such that $\hat{q}_i a_{ij} = \hat{q}_j a_{ji}$ for $i, j \in Q_1$ and $\hat{q}_i a_{ij} = \hat{q}_j a_{ji}$ for $i, j \in Q_2$. Based on the two cases of Theorem 1, under the pinning controller

$$\begin{aligned}
\hat{u}_i & = \sum_{j=1}^{s_1} \hat{q}_i a_{ij} [\zeta \text{sgn}^\alpha(x_j - x_i) + \eta \text{sgn}^\beta(x_j - x_i)] \\
& - b_i \text{sgn}^\gamma(x_i - x_0), i \in Q_1
\end{aligned} \tag{27}$$

all the nodes in Q_1 will reach synchronization in a finite time (or a fixed time), which is denoted as $\mathbf{T}_1$ (or $\mathbf{T}_2$ for the fixed time). After $\mathbf{T}_1$ (or $\mathbf{T}_2$), $e_i = x_i - x_0 \equiv 0, \forall i \in Q_1$. Now, the control inputs for the nodes in Q_2 are updated to

$$\begin{aligned}
\check{u}_i & = \sum_{j=s_1+1}^N \check{q}_i a_{ij} [\zeta \text{sgn}^\alpha(x_j - x_i) + \eta \text{sgn}^\beta(x_j - x_i)] \\
& - \sum_{k=1}^{s_1} \check{q}_i a_{ik} \text{sgn}^\gamma(x_i - x_0), i \in Q_2.
\end{aligned} \tag{28}$$

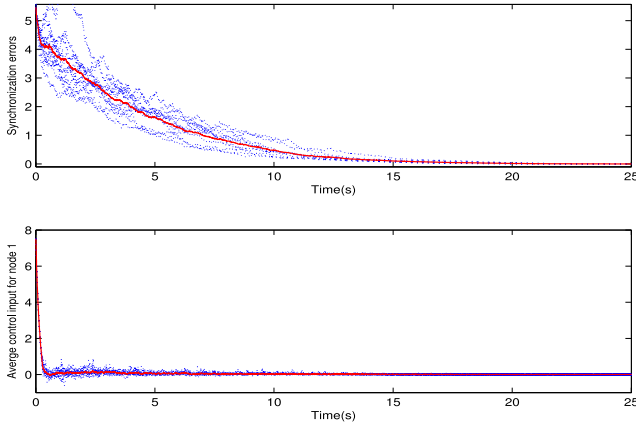


Fig. 1. Individual paths and the average of the finite-time synchronization errors and the control input for node 1.

The rest of the proof can follow a similar line to Theorem 1, where the diagonal matrix $\tilde{\mathbf{b}}$ should update to be

$$\tilde{\mathbf{b}} = \text{diag} \left(\left(\sum_{k=1}^{s_1} \tilde{q}_{s_1+1} a_{s_1+1,k} \right)^{\frac{2}{\alpha+1}}, \dots, \left(\sum_{k=1}^{s_1} \tilde{q}_{N} a_{N,k} \right)^{\frac{2}{\alpha+1}} \right).$$

Then all the nodes in Q_2 will synchronize to x_0 in a finite time or a fixed time. Thus, the proof is completed. ■

Remark 5: The requirement of strongly connected network topology in Theorem 1 is relaxed in Theorem 2, in which some spanning trees are contained. Utilizing the new results obtained in Theorem 1, each strongly connected component of spanning trees can realize FFTS one-by-one and then Theorem 2 is proved by induction. On the other hand, if the considered topology degenerates into an undirected graph, then the corresponding results can be easily obtained just by letting $q_i \equiv 1$ ($i \in \mathcal{N}$), which can be shown in the following example.

IV. NUMERICAL EXAMPLE

In this section, one simulation example is presented to illustrate the effectiveness of theoretical analysis in this brief.

Example: Consider a complex network (1) composed of six nodes (labeled as 1–6). Let the nonzero symmetric elements of the adjacency matrix A be given by $a_{12} = a_{23} = a_{34} = a_{35} = a_{56} = 1$, which is indeed an undirected graph discussed in Remark 5. Let $b_1 = 1$ and the nonlinear function $f(x_i(t)) = 0.07 \tanh(x_i(t))$, $p_i(t, x_i(t)) = 0.07 \cdot \text{diag}(x_i(t))$, then both Assumptions 1 and 2 are satisfied with $\psi_i = \rho = 0.07$ ($i \in \mathcal{N}$), respectively. The initial values are randomly selected as $x_0(0) = (1, 2, 1)^T$, $x_1(0) = -0.3 \cdot \mathbf{1}_3$, $x_2(0) = \mathbf{1}_3$, $x_3(0) = 0.8 \cdot \mathbf{1}_3$, $x_4(0) = 1.3 \cdot \mathbf{1}_3$, $x_5(0) = 0.5 \cdot \mathbf{1}_3$, $x_6(0) = -0.9 \cdot \mathbf{1}_3$, where $\mathbf{1}_3 = (1, 1, 1)^T$.

Case 1: Let power parameters $\alpha = 0.8$, $\beta = 0.4$, and control gains $\zeta = 4$, $\eta = 1$, then $e_\alpha(0) = 1.4083$, $e_\beta(0) = 2.7931$. By the case 1 of Theorem 1, complex network (1) will synchronize to (3) within a finite time and upper-bounds of the settling time is estimated as

$$\begin{aligned} \mathbb{E}\{T_1\} &\leq \min \left\{ \frac{2}{\lambda_3(1-\alpha)} \ln \frac{\bar{\zeta}}{\bar{\zeta} - \lambda_3 e_\alpha(0)}, \frac{2}{\lambda_3(1-\beta)} \ln \frac{\bar{\eta}}{\bar{\eta} - \lambda_3 e_\beta(0)} \right\} \\ &= \min\{58.7088, 19.5519\} = 19.5519. \end{aligned}$$

Case 2: Let power parameters $\alpha = 0.2$, $\beta = 1.6$, and control gains $\zeta = \eta = 2.5$. Then, based on the case 2 of Theorem 1, complex

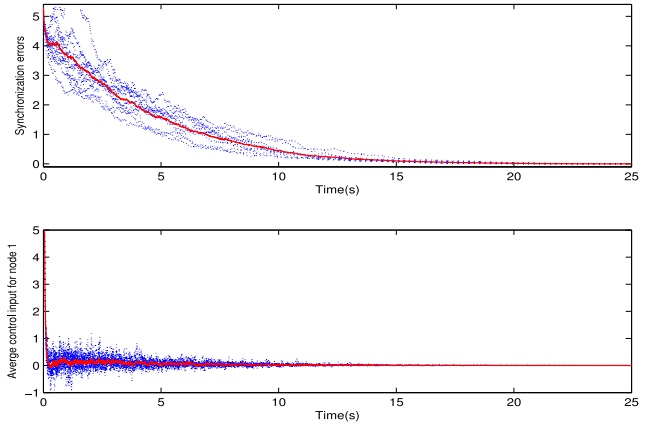


Fig. 2. Individual paths (dotted blue lines) and the average (solid red line) of the fixed-time synchronization errors and the control input for node 1.

network (1) will synchronize to (3) within a fixed time and upper-bounds of the settling time is estimated as

$$\begin{aligned} \mathbb{E}\{T_2\} &\leq \frac{2}{1-\alpha} \frac{1}{\bar{\zeta} - \lambda_3} + \frac{2}{\beta-1} \frac{1}{\bar{\eta} N_\beta - \lambda_3} \\ &= 7.6476 + 14.9881 = 22.6357. \end{aligned}$$

By using the discretization approach in [30], we simulate the standard Brownian motion. Some initial parameters are given as follows: the simulation time $t \in [0, T]$ with $T = 20$, the normally distributed variance $\delta t = T/M$ with $M = 2 \times 10^4$, step size $\Delta t = \varrho \delta t$ with $\varrho = 4$, the number of discretized Brownian paths $p = 10$.

The simulation results are given in Figs. 1 and 2, which depict the evolution of the finite-time and fixed-time synchronization errors ($\|e\|_2 = (\sum_{i=1}^6 \sum_{k=1}^3 |x_{ik} - x_{0k}|^2)^{(1/2)}$) and the control input for node 1 ($u_1 = (\sum_{i=1}^3 u_{1i})/3$), along ten individual paths (dotted blue lines) and the average over ten paths (solid red line).

V. CONCLUSION

In this brief, a unified parameterized pinning controller has been designed to simultaneously realize the FFTS of complex networks with stochastic disturbances. It has been claimed that the synchronization time is either finite or fixed can be decided just by regulating the power parameters in one common controller. One illustrative example has been presented to verify the effectiveness of the new designed results. Further investigations will focus on how to develop less conservative criteria to deal with finite/fixed-time stochastic synchronization under more general communication topologies.

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